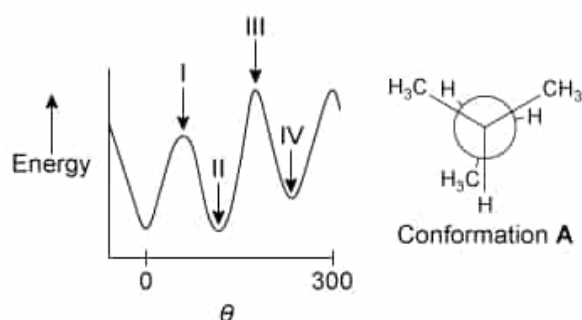


The following plot depicts the torsional energy of 2-methylbutane as it rotates about the C2-C3 bond.



Which point on the energy plot corresponds to the Newman projection shown above?

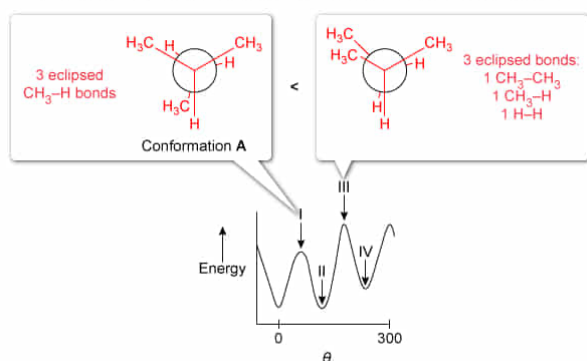
- ✓ ☒ A. I [48%]
☐ B. II [21%]
☐ C. III [20%]
☐ D. IV [9%]

Correct

49% answered correctly

Explanation:

Analysis of a given Newman projection of 2-methylbutane



- Eclipsed conformations are higher in energy than staggered conformations and are represented as peaks.
- Eclipsed CH₃-H bonds are lower in energy than eclipsed CH₃-CH₃ bonds, indicating the given conformation corresponds to I on the energy plot.

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A **conformation** is a specific molecular orientation that results from rotation of groups about a single bond. Since a **sigma bond** results from the head-to-head overlap of orbitals, the relative orientation of the groups can demonstrate up to 360 degrees of variation. A **Newman projection** is a structural representation of molecular conformation. Groups on either the front or back atom can be rotated in increments of 60 degrees to generate a panel of conformations.

The total **torsional energy** of a Newman projection is determined by the presence of conformational features. **Eclipsed** bonds generate torsional energy due to the steric

repulsion of electrons between the aligned groups. Larger alkyl groups generate greater torsional energy for eclipsed bonds. **Staggered** conformations have Newman projection groups oriented at 60° intervals and are lower energy than eclipsed conformations. Alkyl groups with a $\theta = 60^\circ$ (a **gauche conformation**) exhibit modest torsional energy.

The Newman projection given in this question (Conformation **A**) is in an eclipsed conformation. Eclipsed conformations always represent the *peaks* of a torsional energy plot. **Conformational analysis** of 2-methylbutane shows that Conformation **A** has a torsional energy that is different from the other eclipsed conformations of 2-methylbutane due to different pairs of eclipsed bonds. Since none of the eclipsed bonds in Conformation **A** are between methyl groups, Conformation **A** will have a lower relative energy than the other eclipsed Conformations. Together, this analysis demonstrates that Conformation **A** is represented by the *lowest-energy peak* in the torsional energy plot (location **I**).

(Choices B and D) The Newman projection in the question is an eclipsed conformation, which are the *peaks* in a torsional energy plot. Locations **II** and **IV** both represent *valleys* in the plot, which result from staggered conformations.

(Choice C) The Newman projection in the question does not contain any alkyl-alkyl eclipsed bonds that would lead to the largest relative torsional energy (location **III**).

Educational objective:

Conformational analysis evaluates the rotation about a single (sigma) bond. Newman projections provide the ability to analyze the relative torsional energy of conformations.

Time spent: 0 sec

Subject:

Organic Chemistry

QID: 403839

System:

Introduction to Organic Chemistry

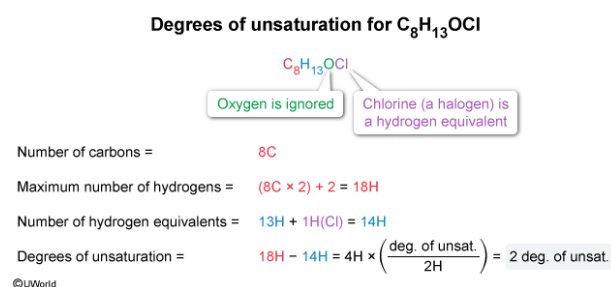
How many degrees of unsaturation are present in a compound with molecular formula $C_8H_{13}OCl$?

- ☐ A. 0 degrees of unsaturation [4%]
☐ B. 1 degrees of unsaturation [25%]
✓ ☒ C. 2 degrees of unsaturation [55%]
☐ D. 3 degrees of unsaturation [13%]

Correct

56% answered correctly

Explanation:



A compound's **molecular formula** describes the type and stoichiometric ratio of elements in a substance. **Isomers** are compounds that share a molecular formula but have different connectivity of atoms. The number of isomers that have a given molecular formula increases exponentially with every additional atom.

A useful approach to narrow the pool of isomers for a molecular formula involves the use of **degrees of unsaturation**. An **unsaturation** is a structural feature that reduces the maximum number of hydrogen atoms in a molecular formula by 2. Each ring or pi bond in a structure counts as one degree of unsaturation. Linear hydrocarbons that are fully **saturated** have a molecular formula of $C_nH_{(2n+2)}$. Other elements (ie, halogens, oxygen, nitrogen) influence the carbon-to-hydrogen ratio and have **additional guidelines** to consider when **calculating degrees of unsaturation**.

Since the molecular formula $C_8H_{13}OCl$ has eight carbon equivalents, a fully saturated molecule would contain $([2 \times 8C] + 2 = 18 \text{ hydrogens})$. Oxygen does not change the carbon to hydrogen ratio and is ignored. Between hydrogen and chlorine, there are 14 hydrogen equivalents. The difference between the saturated and actual number of hydrogens is $18 - 14 = 4$ hydrogen equivalents. Since each degree of unsaturation is 2 hydrogen equivalents, **$C_8H_{13}OCl$ has 2 degrees of unsaturation.**

(Choice A) A saturated molecule with 8 carbon atoms would be expected to have 18 hydrogen atoms $([2 \times 8C] + 2 = 18H)$. The molecule $C_8H_{13}OCl$ has only 14

hydrogen atom equivalents.

(Choice B) A fully saturated hydrocarbon has a molecular formula of $C_nH_{(2n+2)}$.

Forgetting to include the +2 portion of this relationship would result in one fewer calculated degrees of unsaturation.

(Choice D) If oxygen is incorrectly treated as the equivalent of one carbon atom, the fully saturated molecule with nine (8 + 1) carbon atom equivalents would be expected to have 20 hydrogen atoms ($[2 \times 9C] + 2 = 20H$). This would incorrectly result in 3 degrees of unsaturation.

Educational objective:

The degrees of unsaturation are calculated from the information in a molecular formula. Each degree of unsaturation represents the presence of either a ring or a pi bond in the structure of compounds with the molecular formula.

Time spent:0 sec

Subject:

Organic Chemistry

QID:403840

System:

Introduction to Organic Chemistry

Which of the following statements do NOT correctly describe a contributing factor for the increased reactivity of cyclopropane versus cyclohexane?

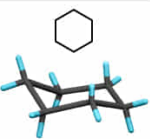
- ☐ A. The carbon-carbon sigma bonds have poor orbital overlap. [27%]
- ☐ B. The bond angle in cyclopropane is compressed to 60° . [14%]
- ✓ ☒ C. Cyclopropane has a lower boiling point. [32%]
- ☐ D. The C-H bonds in cyclopropane are permanently eclipsed. [24%]

Correct

32% answered correctly

Explanation:

Reactivity of cyclopropane and cyclohexane

Compound	 Cyclopropane More reactive	 Cyclohexane
Molecular weight	42.08 g mol ⁻¹	84.16 g mol ⁻¹
Boiling point	-32.7 °C	80.8 °C
Orbital overlap	Not head-to-head (poorer)	Head-to-head
Bond angle	60° (more compressed)	109.5°
C-H relationships	Eclipsed (torsional strain)	Staggered

- Proportional to molecular weight
- Does not impact chemical reactivity

Alkanes are organic molecules that are composed solely of carbon and hydrogen and lack the presence of pi bonds. Alkanes are among the least reactive functional groups due to the nonpolar nature of their C-H and C-C bonds. Cycloalkanes are alkanes that contain a cyclic region of at least three carbon atoms.

The reactivity of cycloalkanes varies with ring size due to differences in **ring strain**. Ring strain encompasses two structural components: **angle strain** and **torsional strain**. Angle strain is present when cycloalkanes have bond angles that deviate significantly from the preferred 109.5° bond angle of a tetrahedral sp^3 carbon atom. Cyclopropane has 60° bond angles and is the cycloalkane with the largest angle strain (**Choice B**). In contrast, the 120° bond angles of planar cyclohexane become 109.5° when it adopts a chair conformation, fully eliminating the angle strain in cyclohexane.

A secondary effect of cyclopropane's compressed bond angles is poor head-to-head overlap of sp^3 orbitals in its carbon-carbon sigma bonds (**Choice A**). Unlike a normal **sigma bond**, where the electron density lies on an axis between the nuclei, cyclopropane carbon-carbon sigma bonds have electron density extended outward from this axis—a feature called **banana bonds**. Banana bonds are weaker than a traditional carbon-carbon sigma bond due to decreased overlap between sp^3 orbitals.

Cyclohexane does not contain banana bonds.

The second component of ring strain is torsional (conformational) strain (which results from eclipsed bonds). The **Newman projection** of cyclopropane is unable to adopt alternate conformations to relieve torsional strain of the **eclipsed C-H bonds (Choice D)**. In contrast, planar cyclohexane adopts one of two **chair conformations**, which allow a lower-energy, staggered conformation in the Newman projection.

The question asks which statement does NOT describe a contributing factor for the increased reactivity of cyclopropane versus cyclohexane. Although cyclopropane has a lower boiling point than cyclohexane due to its smaller molecular weight and smaller amount of London dispersion forces, a compound's **boiling point does not directly impact** its inherent **reactivity**.

Educational objective:

Cyclopropane has a higher relative reactivity than most other alkanes due to its increased ring strain. Ring strain is due to a combination of angle strain and torsional (conformational) strain.

Time spent:0 sec

Subject:

Organic Chemistry

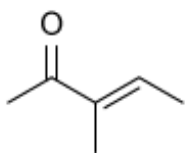
QID:403841

System:

Introduction to Organic Chemistry

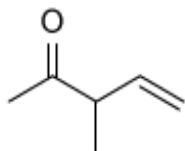
Which of the following line-angle structures correctly represents the structure $\text{CH}_3\text{COCH}(\text{CH}_3)\text{CHCH}_2$?

☐ A.



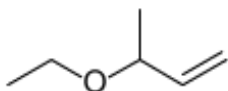
[6%]

✓ ☒ B.



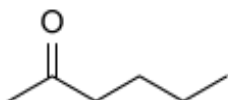
[86%]

☐ C.



[6%]

☐ D.

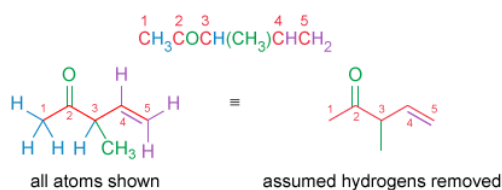


Correct

86% answered correctly

Explanation:

Line-angle and condensed structural formulas



Key

central atoms in the parent chain
hydrogen atoms which generate fully saturated carbons
substituents or branch points on parent chain
hydrogen atoms on unsaturated carbons

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For **condensed structural formulas**, when a molecule contains a branch point or complex substituent, **parentheses** are used to denote the combination of atoms attached at that point. The presence of **nonbonding electrons** and/or multiple bonds may need to be inferred from the structure and adherence to the **octet rule**.

Line-angle (stick) formulas further deemphasize molecular structure. Carbon atoms are implied wherever a line *ends* or *changes direction* (vertex), along with adequate hydrogens to fulfill the octet rule. Multiple bonds and **heteroatoms** (eg, oxygen, nitrogen, sulfur) are *always* shown, and nonbonding electrons may either be depicted or implied.

In this question, there are a total of five central carbon atoms in the parent chain and hydrogen atoms attached to carbons 1, 3, 4, and 5. The oxygen following carbon 2 is implied

to be connected through a double bond, as there are no other atoms following carbon 2 and oxygen prefers to form two bonds. The CH₃ in parentheses indicates that there is a one-carbon branch at carbon 3. To fulfill the octet rule for carbons 4 and 5, a double bond must exist between these two central atoms. Removal of assumed hydrogen atoms results in the structure in **Choice B**.

(Choice A) Carbon 3 cannot participate in a double bond, as it is already attached to four other groups through single bonds. Carbon 5 contains one additional hydrogen beyond what is listed in the given formula.

(Choice C) While an oxygen can potentially be a part of the parent chain (ie, an ether), carbon 2 does not contain any attached hydrogen atoms. Therefore, the oxygen in this question must be attached to carbon 2 through a double bond (ketone) to fulfill the octet rule for carbon 2.

(Choice D) This structure has all carbon atoms in an unbranched parent chain, which does not align with the allocation of hydrogen atoms and use of parentheses in the given condensed formula.

Educational objective:

Condensed structural formulas allow structures to be written within a line of text using parentheses to denote branch points or substituents. Line-angle formulas further simplify molecular structure and greatly deemphasize carbons and hydrogens.

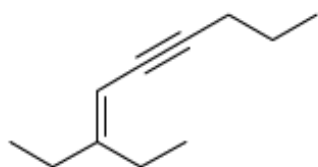
Time spent:0 sec

Subject:
Organic Chemistry

QID:403842

System:
Introduction to Organic Chemistry

What is the IUPAC name for Compound **X** shown below?



Compound **X**

- ☐ A. 7-ethylnon-4-yn-6-ene [16%]
☐ B. 3-ethylnon-5-yn-3-ene [15%]
✓ ☒ C. 3-ethylnon-3-en-5-yne [55%]
☐ D. (3*E*)-3-ethylnon-3-en-5-yne [12%]

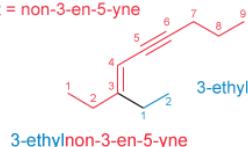
Correct

56% answered correctly

Explanation:

IUPAC name of 3-ethylnon-3-en-5-yne

Parent = non-3-en-5-yne



3-ethylnon-3-en-5-yne

Alkenes and alkynes have the same IUPAC priority, with alkenes listed before alkynes due to alphabetical order of the parent chain modifier. The parent chain is numbered to give the alkene and alkyne functional groups the lowest numbering.

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IUPAC provides a systematic method of naming organic compounds. A compound's **parent chain** is the longest continuous chain of carbon atoms that contains the greatest number of **highest-priority functional groups**; it is numbered to provide the highest-priority functional groups the lowest possible number. The parent chain name includes a **prefix** denoting the number of carbon atoms in the chain, a middle portion denoting the presence of select pi-bond functional groups, and a suffix denoting the quantity and type of highest-priority functional group.

Lower-priority functional groups are named as **substituents**, which include a numerical position of attachment to the parent chain. Substituent chains are numbered from the point of attachment to the parent chain and named using a prefix to indicate the carbon chain length and a *-yl* suffix.

The parent chain in this molecule contains nine carbon atoms (*non-* prefix), an alkene, and an alkyne. Alkenes and alkynes have the same priority level for IUPAC nomenclature; in cases where a parent chain contains both types of functional groups, the alkene (*-en-*) comes before alkyne (*-yn-*) due to alphabetical order. The parent chain is numbered to provide the lowest

number for the alkene and alkyne functional groups: non-3-en-5-yne. The substituent located at position 3 (ethyl) leads to the IUPAC name of **3-ethyl-non-3-en-5-yne**.

(Choice A) The parent chain is numbered to provide the pi-bond functional groups the lowest possible numbering. 7-Ethylnon-4-yn-6-ene incorrectly numbers the parent chain from the end closer to the alkyne, and it also incorrectly lists the alkyne first in the name.

(Choice B) Alkenes and alkynes have the same priority for IUPAC nomenclature and are listed alphabetically within the parent chain. The name 3-ethylnon-5-yn-3-ene incorrectly lists the alkyne (-yn-) *before* the alkene (-en-).

(Choice D) An *E/Z designation* for an alkene is possible only if each half of the alkene contains different functional groups. (3*E*)-3-ethylnon-3-en-5-yne incorrectly includes a 3*E* designation when C3 has two equivalent ethyl groups.

Educational objective:

IUPAC provides a systematic method to generate a name for organic compounds. Alkenes and alkynes have the same priority and are listed alphabetically within the parent chain name.

Time spent:0 sec

Subject:
Organic Chemistry

QID:403843

System:
Introduction to Organic Chemistry